

Paper B: corrections to the manuscript (master plan)

Against the author’s draft committed at e7b997e3, 30 September 2026

Each row sets the current manuscript text (left) against the proposed replacement (right), both typeset as they would appear. BEFORE text is cut from the committed manuscript by script, so it matches the manuscript exactly. Under each entry, **Why** gives the error and **Evidence** the Lean or sympy record or page that settles it (audit items refer to `notes/citation_audit.md`). Entry C.15 (inside C.11.9) reconciles Section 3 with `notes/positioning_economics.tex`; the abstract (C.1) leads with the identification point for the same reason. Cross-references print as their label (e.g. PROP:IMM); footnotes print inline. **Nothing here is applied.** The author approves entries by number; approved entries are applied to the manuscript, compiled and committed. Every AFTER text passes the mechanical checks of `notes/writing_discipline.md` (no colons, no dashes doing a sentence’s work, “sequence” for reading order, the fixed adoption terminology, length within 80–120% of the draft except for deletions, citation fixes and the removed duplicate); the generator refuses to build otherwise.

Section C: corrections to the current draft

Part	BEFORE (current manuscript)	AFTER (proposed)
C.1	Abstract, sentences 2-4	
	Considering a case where evaluators reading two correlated soft cues compare their beliefs as well as decisions against a sequence-free Bayesian benchmark, the current papers asks which of the beliefs and decision statistics register the resulting order effect. Considering the empirical limitations in addition to theoretical underpinnings, the paper shows that an arbitrary statistic is an order of magnitude closer to the benchmark when two cue-sequences displace a belief within a two-dimensional plane. This results in measures of the believed association and the average value lost through decisions differing from benchmark staying close to the benchmark while the share of the population affected by the sequence differing from the benchmark.	Considering evaluators who read two correlated soft cues and comparing their beliefs and decisions with a sequence-free Bayesian benchmark, the paper asks which statistics of those beliefs and decisions register the resulting sequence effect. Whether a statistic registers it is a question of identification rather than of estimation, since the two reading sequences move a belief in only two directions, and a statistic insensitive to both carries no first-order signal in the population itself, however many evaluators are sampled. The believed association between the attributes and the average value lost through decisions that differ from the benchmark are of that kind, while the marginal probabilities and the share of the population whose decision the sequence changes depart from the benchmark at first order.
Why. “An arbitrary statistic is an order of magnitude closer to the benchmark” is false as stated; only the protected statistics are (Proposition PRO; plan 0.A made the same point). The two-dimensional plane is the mechanism, not the finding; for the Journal of Mathematical Economics the abstract leads with the economic content, that a null on a belief audit is an identification failure and not an estimation one (the point of <code>notes/positioning_economics.tex</code> , already in the conclusion at MS ~950). Also “the current papers asks”, and “This results in measures ... staying ... while the share ... differing” has no main verb. “Considering the empirical limitations in addition to theoretical underpinnings” is cut because no empirical limitation is named. Evidence. DEC and LOS (second order), DRF and SHR (first order), PRO (the plane and the classification). All in the Lean and sympy suites.		
C.2	Introduction, paragraph 1, sentences 2 and 4	
1/2	The sequence-dependence of Jeffrey conditioning for soft evidence has attracted much debate (Diaconis and Zabell, 1982; Hawthorne, 2004) but the mechanism of how sequence dependence register in observed statistics remains unclear (Zhao and Osherson, 2010).	The sequence dependence of Jeffrey conditioning on soft evidence has attracted debate (Diaconis and Zabell, 1982; Hawthorne, 2004; Zhao and Osherson, 2010), but how that dependence registers in the statistics an observer reads has received little attention.

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2/2	It finds that the share of population affected by evidence sequence deviates from a sequence-invariant benchmark – but the believed association between attributes and the total loss due to sequence effects do not.	It finds that the share of the population whose decision the sequence changes departs from a sequence-invariant benchmark at first order, while the believed association between attributes and the surplus-weighted loss depart from it only at second order.
Why. Zhao-Osherson do not say the mechanism “remains unclear”; they say doubts about order dependence “disappear on closer inspection of the evidential weight of probability judgements” (p. 291). Cited for the debate, as the author agreed. Jeffrey himself says non-commutativity “is as it should be” (1983, pp. 182-183; audit J1) and Diaconis-Zabell that it “is not a real problem” (Remark 2, p. 827; M24); Hawthorne calls the order effect “very troubling” (p. 99). Sentence 4: the loss is not free of the sequence effect, it carries it at second order (LOS), and “– but” is a dash doing a sentence’s work. Evidence. verify_jeffrey.md J1; verify_kinematics.md; verify_hawthorne_weisberg.md; ZO p. 291 read 2026-09-30; Theorem LOS, Proposition SHR.		
C.3 Introduction, paragraph 2 (the literature paragraph) and its footnote		
	The discussion on Jeffrey (Jeffrey, 1983) conditioning’s dependence on arrival sequence have focused more often on whether the commutation criterion is exactly satisfied or not but not on how measurable it is in observed data. The impression literature (Asch, 1946) in fact often models an impression as a single score, one attribute at a time, so it has only one statistic to read. While the characterisation of how two sequences could lead to the same belief Diaconis and Zabell (1982); Hawthorne (2004) does consider a joint belief over attributes, the order dependence is more likely to be treated as a defect to repair <i>[Footnote: Hawthorne (2004) repairs this by letting a cue multiply the old belief instead of replacing it, and under his order-free variants the sequence effect disappears.]</i> rather than as something to measure. While the changing of input from a replacing credence to a multiplying factor settles whether beliefs depend on sequence or not, it hardly addresses the empirical issue that an observer cannot tell whether impressions replace prior belief or adjust it.	Discussion of how Jeffrey conditioning (Jeffrey, 1983) depends on arrival sequence has more often asked whether the commutation criterion is exactly satisfied than how measurable a failure of it is in observed data. The belief-adjustment literature models an impression as a single score (Hogarth and Einhorn, 1992), and the founding multi-trait study reports its results trait by trait (Asch, 1946), so both read marginals only. The characterisation of when two sequences lead to the same belief does consider a joint belief over attributes (Diaconis and Zabell, 1982), but where order dependence has been treated as a problem it has been treated as a defect to repair <i>[Footnote: Hawthorne (2004) offers factor-based alternatives in which a cue supplies a normed-likelihood or likelihood-ratio factor instead of a new probability. Under these, updates on distinct attributes commute, and in his Basis-Commuting Version all updates do.]</i> rather than as something to measure. Replacing a credence that resets the prior marginal by a factor that multiplies it settles whether beliefs depend on sequence, but it leaves open the empirical question of whether an impression replaces a prior belief or adjusts it.
Why. (i) Asch did not model an impression as a single score; Experiment VI reports eighteen traits under two orders (audit A28). His published tables give per-trait percentages, i.e. marginals. (ii) Diaconis-Zabell are grouped with the defect view, but they hold that non-commutativity “is not a real problem” (M24). (iii) The footnote says Hawthorne “repairs” the defect and that the effect disappears; he offers alternatives, and order-freedom holds across distinct bases, fully only in his Basis-Commuting Version (M19). (iv) \citet inside a sentence where \citep is meant; “have focused” agreement. Evidence. Asch1946 record (Asch.lean, Table 7); DiaconisZabell.lean and README; Hawthorne.lean (<code>extUpdate_basisCommuting</code>, <code>factorUpdate_comm</code>), Sections 6-8.		
C.4 Introduction, the two-channel paragraphs (appear twice)		

Part	BEFORE (current manuscript)	AFTER (proposed)
	More specifically, arrival-sequence dependence can enter through two channels, an as- sociation channel, in which one cue changes what the other implies and which exists only when the attributes are believed related, and a position channel, in which the observer weights the later cue less whatever the attributes are (Hogarth and Einhorn, 1992; Asch, 1946). Unlike the position channel, where the later cue is adopted only in part and every marginal registers the sequence even when the attributes are believed unrelated, the association channel is one where the sequence registers unevenly across statistics due to each impression being adopted in full (Diaconis and Zabell, 1982) so that the position channel is absent. More specifically, arrival-sequence dependence can enter through two channels, an association channel, in which one cue changes what the other implies and which exists only when the attributes are believed related, and a position channel, in which the observer weights the later cue less whatever the attributes are (Hogarth and Einhorn, 1992; Asch, 1946). Which channels operate is fixed by how fully the later cue is adopted. When it is adopted only in part, the position channel operates and every marginal registers the sequence even when the attributes are believed unrelated. When each impression is adopted in full (Diaconis and Zabell, 1982), the position channel is absent and the association channel is the only one – so that the sequence registers unevenly across statistics. The association channel is the setting that the current paper explores. In the two-attribute setting, the paper therefore considers three coordinates of a belief – two marginal probabilities and the cross-attribute association – and explains how a difference from order-free benchmark below the observer’s precision is invisible in the statistic read by the observer. The question that paper addresses is not simply whether the two sequences agree but by how much they disagree in an arbitrary statistic. The answer the paper finds and details is that whereas the marginal probabilities and decisions thereof carry the difference at first order in the prior covariance, the believed association and related statistics carry it only at second order.	More specifically, arrival-sequence dependence can enter through two channels. In the association channel one cue changes what the other implies, and the channel exists only when the attributes are believed related. In the position channel the weight a cue receives depends on where in the sequence it arrives, whatever the attributes are. When the first cue is adopted in full and later cues only in part the earlier impression is favoured, and when every cue is adopted only in part the later one is (Hogarth and Einhorn, 1992). In the model studied here, which channels operate is fixed by how fully the later cue is adopted. When it is adopted only in part, the position channel operates and every marginal registers the sequence even when the attributes are believed unrelated. When each impression is adopted in full, as in the successive updating of Diaconis and Zabell (1982), the position channel is absent and the association channel is the only one, so the sequence registers unevenly across statistics. That full-adoption setting is the one the current paper explores. A belief in the two-attribute setting has three coordinates, the two marginal probabilities and the cross-attribute association, and a departure from the sequence-free benchmark smaller than the observer’s precision is invisible in the statistic read. The question is therefore not only whether the two sequences agree but by how much they disagree in a given statistic. The marginal probabilities and the decisions made from them carry the difference at first order in the prior covariance, while the believed association and the statistics that move with it carry it only at second order.
Why. (i) The paragraph appears twice; the first copy is an earlier version and lacks its final period. The AFTER replaces both copies. (ii) “the observer weights the later cue less” is cited to Hogarth-Einhorn and Asch, but Hogarth-Einhorn’s step-by-step partial adjustment predicts recency (Appendix B) and their primacy comes from anchoring on the first item (End-of-Sequence, Eq. 8); Asch denies that “sheer temporal position” matters (p. 272). Asch is dropped (audit P10-P12). (iii) “Which channels operate is fixed by how fully the later cue is adopted” is true of this model but not in general (under base-rate neglect the later cue is adopted in full and a position channel still operates), so it is scoped to “the model studied here”. (iv) Full adoption is Diaconis-Zabell’s setting, not their premise. Optional, if a published instance of a position channel with likelihood inputs is wanted: add after the Hogarth-Einhorn sentence “Under base-rate neglect, where each posterior becomes the prior for the next cue, the earlier cue is discounted \citep{BenjaminBodohCreedRabin2019}.” (needs the bib entry in C.12). Evidence. HogarthEinhorn.lean (appB_recency, eq8_estimation_first_dominates, oneSided_primacy_iff); BenjaminBodohCreedRabin.lean; verify_interior_omega.py rows 1-2; Proposition ADJ endpoints (Anchoring.lean).		
C.5 Introduction, hiring-panel paragraph, last sentence		
	In demonstrating how the sequence-dependence of certain statistics can be invisible to an observer, the panel example also clarifies that the invisibility does not depend on the extent to which a later impression erases the earlier one (the amnesic updating concern in the literature).	In the panel example the odds ratio between the two traits is the same in both reading sequences however far a later impression erases the earlier one, since each update rescales rows or columns of the belief (Lemma LEM:SEP), whereas the cross-product association is protected in that way only under full adoption (Section SEC:SCOPE).
Why. False as written for the cross-product association, which is the paper’s assoc. Under partial adoption its sequence effect is first order in c, with coefficient $(1-\omega)$ times a nonzero term, and the pooled association is nonzero already at $c = 0$. Only the odds ratio is identical across sequences for every ω. The commented-out line below the paragraph states the odds-ratio version. The AFTER points to Section 6, where plan 6.B carries the interior-omega numbers; apply 6.B with it. Evidence. sympy/verify_interior_omega.py checks (2), (3), (4); Lemma SEP (LemmaSEP.lean, general N).		

Part	BEFORE (current manuscript)	AFTER (proposed)
C.6 Introduction, premise paragraph, from “Read as a Bayes factor”		
	Read as a Bayes factor, the credential carries a likelihood ratio stating how much more probable it is under competence than in its absence, and that ratio multiplies whatever belief it meets. A Bayes factor, however, requires the probability of the same credential for a candidate who is not competent, and an impression does not supply that counterfactual. As a mechanism that holds conditionals fixed, Jeffrey updating is the unique coherent revision for credence delivered on the cue’s partition and is equivalent to making the minimal change of the prior consistent with the delivered marginal (Diaconis and Zabell, 1982)	Read as a Bayes factor, the credential carries the ratio of new to old odds on competence, which multiplies whatever belief it meets. The two readings agree on a single cue (Proposition PROP:IMM) and differ in what stays fixed when the cue meets another prior. The panelist’s own impression is naturally a credence, whereas a factor suits evidence reported by someone else, whose report mixes the evidence with that person’s prior (Jeffrey, 2004, ch. 3). Jeffrey updating holds fixed the conditionals given the cue’s partition, and its posterior is the unique belief with the delivered marginal closest to the prior in Kullback–Leibler divergence (Diaconis and Zabell, 1982, Theorem 5.1)
Why. (i) “A Bayes factor requires the probability of the same credential for a candidate who is not competent” is contradicted by Jeffrey’s own definition, new odds over old odds, which needs no likelihood (2002 draft of Jeffrey 2004, ch. 3; audit J2). Jeffrey draws the line between the readings by provenance instead (own experience gives credences, others’ reports should be converted to factors), which supports the paper’s modelling choice. (ii) “the unique coherent revision” is not in Diaconis-Zabell; what they prove is the unique Kullback-Leibler (and Hellinger) minimiser (M22). “equivalent to making the minimal change” without naming the distance is also loose (not unique in variation distance). The footnote after the citation is unchanged. Needs the Jeffrey2004 bib entry (C.12); the Drive copy is the 2002 draft, so cite the chapter, not a page. Evidence. verify_jeffrey.md J2; DiaconisZabell.lean (thm51_KL_eq_iff, jcond_iff_jeffrey); PropIMM.lean (propIMM_single_cue).		
C.7 Introduction, terminology paragraph, Domotor sentence and two typos		
1/2	The sequence-dependence of marginals follows easily – since the likelihood a delivered credence implies must be read against the marginal in force when the cue arrives (Domotor, 1980).	The sequence dependence of the marginals follows directly, since a delivered credence is taken against the marginal in force when the cue arrives, which the first cue moves when $c \neq 0$ (Proposition PROP:DIV).
2/2	To avoid a clash in terminology we use “order” to describe proximity with a sequence-free Bayesian benchmark and “sequence” to denote the sequence in which evidence is read.	To avoid a clash in terminology we use “order” to describe proximity with a sequence-free Bayesian benchmark and “sequence” to denote the sequence in which evidence is read.
Why. Domotor never mentions likelihoods or marginals; he supports only bare non-commutativity, and treats it as a defect of Jeffrey machines (p. 395), the opposite lean (audit M1). The mechanism is the paper’s own (Proposition DIV). If a Domotor citation is wanted, it fits the literature paragraph as a source of the commutativity demand that Jeffrey (1983, pp. 182-183) rejects. Typos: “bechmark”, sequence closed with backquotes. Evidence. Domotor.lean (field_eq_jeffrey_embed, embed_depends_on_state); Cripps.lean (composite_AB_eq_bayes: the B-marginal moves by $c(q_0 - \alpha) / (\alpha(1 - \alpha))$).		
C.8 Introduction, key-finding paragraph, last two sentences		
	The implication for evaluation is that and audit’s answer depends on which statistic is aggregated rather than on how many evaluators are averaged over. One consequence, is that the instruments most naturally trusted for detecting stereotype—belief audits asking how attributes are thought to go together—may not even be detectable.	The implication for evaluation is that an audit’s answer depends on which statistic is aggregated rather than on how many evaluators are averaged over. One consequence is that belief audits asking how attributes are thought to go together, the instruments most naturally trusted for detecting stereotype, cannot detect the sequence effect at first order.
Why. Typos (“and audit’s”, “One consequence, is”), and “the instruments ... may not even be detectable” says the instruments are undetectable; it is the sequence effect that belief audits cannot detect at first order. Evidence. Propositions DEC, PRO.		
C.9 Setup 2.1, the prior paragraph and the sequence/weight sentences		
1/3	The three numbers that fix P are described as Δ^3 – specified using the marginals $P(A=0) = \alpha$, $P(B=0) = \beta$ and the prior covariance c between the attributes:	The prior is a point of Δ^3 fixed by three numbers, the marginals $P(A=0) = \alpha$ and $P(B=0) = \beta$ and the prior covariance c between the attributes,
2/3	Simply put, the marginals $\alpha = P(A=0)$ and $\beta = P(B=0)$ denotes the shares of applicants the panelist believes to lack competence and to lack trustworthiness and the prior covariance $c = \text{assoc}(P)$ denotes how strongly the panelist believes the	Simply put, the marginals $\alpha = P(A=0)$ and $\beta = P(B=0)$ denote the shares of applicants the panelist believes to lack competence and to lack trustworthiness, and the prior covariance $c = \text{assoc}(P)$ denotes how strongly the panelist believes the

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3/3	To address the issues in amnesic updating debate, we also consider an adoption weight $\omega \in [0, 1]$ – which denotes how far the letter displaces what the credential had already implied about trustworthiness. At $\omega = 1$ the letter sets the rating outright, and at $\omega = 0$ the credential’s implication is never moved.	To address the debate over whether a later impression should be adopted in full (Hawthorne, 2004), we also consider an adoption weight $\omega \in [0, 1]$, which denotes how far the letter displaces what the credential had already implied about trustworthiness. At $\omega = 1$ the letter sets the rating outright, and at $\omega = 0$ the credential’s implication is never moved.
Why. Grammar ("described as Δ^3 –“, "denotes“ for two subjects). "Amnesic" is used only where Hawthorne is quoted (writing_discipline.md 6). Dependency: omega is defined here but used nowhere else in the manuscript until plan 5.B (Proposition ADJ) is applied; either apply 5.B or move this sentence into 5.B. The Frechet-Hoeffding bounds in the new text are correct (checked cell by cell). Evidence. Bounds: nonnegativity of the four cells of P; Hawthorne pp. 98-99.		
C.10 Setup 2.2, the invariance condition and the justification of the rule		
1/2	holding the conditionals fixed—the invariance condition (Jeffrey, 1983, ch. 11).	holding the conditionals given the cue’s partition fixed, the invariance condition (Jeffrey, 2004, ch. 3), which Jeffrey (1983, ch. 11) expresses by saying that the change originates in the partition.
2/2	The axiomatic grounding is twofold. The partition satisfying the invariance condition is the minimal sufficient statistic for revising the prior to any candidate posterior on that partition (Diaconis and Zabell, 1982), and the rule is the minimal revision of the prior consistent with the delivered marginal, in the I -divergence sense (Diaconis and Zabell, 1982).	Two results of Diaconis and Zabell (1982) support the rule. The invariance condition makes the cue’s partition sufficient for the prior and posterior, and minimal whenever the delivered credence differs from the prior marginal (their Theorem 2.2). The rule is also the unique minimiser of the I -divergence from the prior among all beliefs with the delivered marginal (their Theorem 5.1).
Why. (i) Chapter 11 of Jeffrey (1983) never uses “invariance”; it says the change “originates in” the partition (pp. 168, 174). “Invariance” is the 2004 book’s term (audit J4). (ii) “The partition satisfying the invariance condition is the minimal sufficient statistic for revising the prior to any candidate posterior” misstates D-Z Theorem 2.2: sufficiency is for one pair $\{P, P^*\}$, every refinement of such a partition also qualifies, and the minimal one is the likelihood-ratio partition (M23; Jeffrey p. 174 also allows “a certain latitude in the choice” of partition, J3). (iii) D-Z give no axioms; they call this “mechanical updating”. The summation index is x because ω is now the adoption weight. Evidence. DiaconisZabell.lean (thm22_lr_minimal, jcond_of_refines, lr_jeffrey_iff, thm51_KL_eq_iff); verify_jeffrey.md J3-J4.		
C.11 Related literature, the paragraph from BHW/Banerjee to Becker and Good-Mittal		
1/9	In Banerjee (1992), the sequence persists because a coarse public action fails to be a sufficient statistic for private information. In Bikhchandani et al. (1992), on the other hand, a cascade is an action that conveys no private signal.	In both, the equilibrium action is not a sufficient statistic for private information, being binary in Bikhchandani et al. (1992) and a non-invertible choice rule in Banerjee (1992), so once a cascade or herd forms, later actions convey no private signal.
2/9	One similarity with the literature, however, is that the sequence effect in (Bikhchandani et al., 1992; Banerjee, 1992) lives on the analogue of the marginals—which remain so in the current paper as well.	One analogy remains. Those models have a single state, so their sequence effect falls on a belief about one variable, the counterpart of a marginal, and here too the marginals carry the sequence effect.
3/9	A related normative literature evaluates group belief formation by whether aggregation commutes with updating—a design criterion for the pooling rule (Dietrich, 2021). While geometric pooling (Dietrich, 2021, Def. 1) satisfies the criterion, the population mean studied in the current paper (i.e. the average posterior across evaluators who differ only in the sequence they meet the two cues) is the outcome of linear pooling and generically fails the criterion. As further sections elaborate, this is consistent with the first-order divergence of every statistic outside the protected class of Proposition PROP:PRO.	A related normative literature evaluates group belief formation by whether aggregation commutes with updating, a criterion that geometric pooling meets and linear pooling generally does not (Dietrich, 2021, Theorem 2). The criterion does not discriminate here. Evaluators who share a prior and differ only in reading sequence do not hold conditionalisations of that prior on a common event, and the linear and geometric averages of their posteriors differ only at $\mathcal{O}(c^2)$. The first-order divergence of every statistic outside the protected class of Proposition PROP:PRO is therefore a property of Jeffrey updating, not of linear pooling.

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4/9	Several alternatives to Bayesian updating have been used in the decision theory literature to explain whether sequential conditioning is sequence-independent or not. For example, Epstein (2006) makes the updating rule subjective, and Ortoleva (2012) axiomatises departures triggered by unexpected news. Cripps (2021) shows that symmetry and divisibility jointly force sequence-independence, so the correlated two-cue composite of Proposition PROP:DIV fails divisibility—and, of the four axioms, only divisibility. <i>[Footnote: Cripps’s axioms govern rules on fixed-likelihood experiments; the statement here is behavioural—the map from prior and delivered credences to final belief, held against the sequence-invariance that his Symmetry and Divisibility axioms jointly force.]</i>	Decision theory has axiomatised several alternatives to Bayesian updating, each for a single piece of news. Epstein (2006) makes the updating rule, and not only the prior, subjective, and Ortoleva (2012) axiomatises departures triggered by unexpected news. Cripps (2021) treats the sequence of signals, and his Symmetry and Divisibility axioms imply that two conditionally independent signals give the same posterior in either sequence. Read with each cue as its likelihood matched to the prior (Proposition PROP:IMM), the composite of Proposition PROP:DIV is a sequence of Bayes updates satisfying all four of his axioms, and it depends on the sequence because the second likelihood is matched to the intermediate belief, so the two sequences process different experiments.
5/9	While the literature restores sequence-invariance for sequential Jeffrey updating by changing how successive inputs are pooled (Pettigrew and Weisberg, 2025),	While Pettigrew and Weisberg (2025) restore sequence-invariance by pooling the prior with each new input multiplicatively before the Jeffrey step,
6/9	The paper borrows the evaluator-with-binary-attributes frame from the statistical-discrimination lineage (Phelps, 1972; Arrow, 1973).	The paper borrows the evaluator-with-binary-attribute frame from the statistical-discrimination lineage (Arrow, 1973; Coate and Loury, 1993).
7/9	instead of stereotypes as representativeness-distortion or selective recall (Bordalo et al., 2016), the sense of stereotype in the current paper is that of a believed cross-attribute association produced by coherent updating on impoverished input, not by a distortion of memory or sampling.	instead of stereotypes as the selective recall of representative types (Bordalo et al., 2016), whose exaggerated associations arise across groups, the stereotype in the current paper is a believed cross-attribute association held by one evaluator and produced by coherent updating on impoverished input, not by a distortion of memory.
8/9	The lineage distinction the concluding section trades on—protection intrinsic to the arithmetic of aggregation, rather than imposed by an enforced constraint on behaviour—goes back at least as far as Becker (1962).	<i>(sentence deleted)</i>
9/9	It is worth contrasting the phenomenon explored in the current paper against the amalgamation paradox (Good and Mittal, 1987), where a real effect present in every subpopulation is erased or reversed by a confound in how the subpopulations are weighted together. Whereas the amalgamation paradox erases a real effect by combining subpopulations under confounded weights, our decision-space result shows the same kind of erasure can occur within a single population – where a first-order share of individuals who are individually affected contributes only a second-order loss in aggregate (Theorem THM:LOS).	It is worth contrasting the phenomenon explored here with the amalgamation paradox (Good and Mittal, 1987), in which a measure of association on a combined table lies outside the range of the same measure on its subtables, because treatment is allocated unevenly across the subpopulations, even when they are equally large. That paradox arises from combining subpopulations. The decision-space result here shows a comparable loss of visibility within a single population, where a first-order share of individuals who are individually affected contributes only a second-order loss in aggregate (Theorem THM:LOS). The audit perspective has a precedent in the economics of discrimination. Heckman (1998) showed that audit-pair estimates of discrimination rest on an assumption about unobserved productivity that “nothing guarantees” (p. 109), so that such an audit “can find discrimination when in fact none exists; it can also disguise discrimination when it is present” (p. 102), and Bohren et al. (2019) identify the source of discrimination by conditioning on the history of evaluations, since in static data, as they note, different sources generate the same patterns of observable behaviour. The obstruction here is of the same kind and arises on the belief side of an audit. The first-order signal in a protected statistic is zero in the population itself, so its silence is a failure of identification and not of estimation, which no sample size repairs, and the identifying variable is the reading sequence, which pooled data discard.

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Why. One entry per sentence group, in manuscript order. BHW/Banerjee: “coarse” belongs to BHW’s binary action, not Banerjee’s continuum; the two share one mechanism, so “on the other hand” goes; the marginals point is the paper’s analogy (M2, M3). Dietrich: Def. 1 is preference aggregation, and with a common prior linear pooling passes the criterion wherever it applies (M4, M5). Epstein/Ortoleva/Cripps (bold sentence): the framing is unsupported for Epstein and Ortoleva; the Cripps claim is false, since under Proposition IMM’s reading the composite satisfies all four axioms, and the footnote goes with it (M6-M9). Pettigrew-Weisberg pool the prior with each input, not successive inputs (M25). Phelps has no binary attribute, so he is dropped from this sentence and can be re-cited in plan 1.C for statistical discrimination in general (M10). BCGS name one mechanism, not two, and “sampling” is not theirs (M12). Becker: the concluding section never draws the distinction, and Becker’s own mechanism is averaging plus a budget constraint, the opposite side of it (“Our statement goes beyond arithmetic”, p. 7; M13); the AFTER deletes the sentence, and an alternative that keeps Becker is in literature/becker1962/README.md. Good-Mittal: their paradox covers amplification and effects created from none, and its cause is uneven allocation, not weighting by population shares (M14). Evidence. BHW.lean, Banerjee.lean, Dietrich.lean, Epstein.lean, Ortoleva.lean (via the 2024 survey), Cripps.lean (<code>order_invariance</code>, <code>composite_AB_eq_bayes</code>), PettigrewWeisberg.lean, Phelps.lean, Arrow.lean, CoateLoury.lean, BCGS.lean, Becker.lean, GoodMittal.lean (<code>piR_amalg_general</code>, <code>equalSize_reversal</code>). The appended paragraph (C.15) carries the identification point of notes/positioning_economics.tex and plan 3.C/3.D into Section 3, with audit P17a (no sample-size claim attributed to Heckman; that clause is the paper’s own consequence of Proposition PRO), H2 (“nothing guarantees” is p. 109), H3 (“can find discrimination” is p. 102) and B37 (BIR credit “different sources” to Fang-Moro, so it is paraphrased) applied. Needs Heckman1998 and Bohren2019 in bibliography.bib (plan B.A).		
C.13 Section 4, the Foster-Greer-Thorbecke sentence		
	This is the standard incidence-versus-intensity pairing of the measurement literature, where a headcount (a count of those past a threshold) and a mean shortfall (average of their distances from it) are first two members of a single parametrised family of indices (Foster et al., 1984).	The two decision statistics pair incidence with intensity, as do the first two poverty measures of Foster et al. (1984), the headcount ratio, the share below a threshold, and the average shortfall below it normalised by the threshold, to which those above it contribute zero.
Why. P_0 is the headcount <i>ratio</i>, a share, and $P_1 = (1/n) \sum g_i / z$ is normalised by the threshold and averaged over the whole population; “average of their distances” is the income-gap measure, which is not in the family (M15). FGT do not use “incidence/intensity”; the AFTER does not attribute it to them. Evidence. FGT.lean (<code>P0_eq_H</code>, <code>P1_eq_popMean_normGap</code>, <code>I_not_decomposable</code>).		
C.14 Appendix, proof of Theorem LOS, the two Tao citations		
1/2	Tonelli’s theorem (Tao, 2011, Corollary 1.7.23) gives	Tonelli’s theorem (Tao, 2011, Theorem 1.7.15), applicable since the integrand is non-negative, gives
2/2	As μ is finite, continuity from above (Tao, 2011, §1.4) yields	As μ is finite and B_c decreases as $c \downarrow 0$, continuity from above (Tao, 2011, Exercise 1.4.23(iii)), applied along any sequence $c_n \downarrow 0$, yields
Why. Corollary 1.7.23 is Fubini-Tonelli; for the nonnegative integrand Tonelli (Theorem 1.7.15, or 1.7.18 for complete measures) is the exact reference. Continuity from above is Exercise 1.4.23(iii), stated for sequences; the proof takes c to 0 over a continuum (M16). Numbering is from the author’s preprint of Tao; confirm against the printed book. Evidence. Tao.lean (<code>thm_1_7_15</code>, <code>ex_1_4_23_iii</code>, <code>measure_band_tendsto_zero</code>, <code>los_step4</code>).		

C.12: bibliography entries the AFTER texts need

Hawthorne2004 and Zhao0sherson2010 are already in `bibliography.bib`. If C.4’s optional sentence, C.6, C.10 or C.15 are approved, four more are needed (Jeffrey 2004, Benjamin et al. 2019, Heckman 1998, Bohren et al. 2019); they are in `notes/manuscript_corrections_extra.bib` so that this document renders them, and move to `bibliography.bib` on approval. The Drive copy of Jeffrey (2004) is the November 2002 draft, so C.6 and C.10 cite the chapter only.

Section D: status of the older plan entries (`notes/manuscript_change_plan.md`)

Entry	Subject	Status and what it needs
0.A	Abstract	Superseded by the author’s rewrite and C.1.

Entry	Subject	Status and what it needs
1.A, 1.B, 1.D, 1.E	Introduction	Superseded by the author’s rewrite; what survives is corrected in C.2-C.8. The ORD sentence of 1.E is still unapplied and presupposes 5.B.
1.A2	Two-channel prelude	Applied by the author; corrected by C.4.
1.C	Why sequence matters beyond one judgement	Needs audit P4 (“prior beliefs about the groups”) and D13 (BIR’s impartial type) before applying.
2.A, 2.C	What an impression is; soft means $0 < q < 1$	Still applicable as written.
2.B	Worked example	Still applicable; every number is in <code>verify_example.py</code> .
3.A	Jeffrey-commutativity literature	Needs D1/P3 (Doring normative, Hawthorne psychological), P2 (Doring’s remedy is one Jeffrey update), M21 (“two ends of his taxonomy”; NL denominator), M24 (D-Z on the paper’s side).
3.B	Asch	Needs P13 (“no prior association is elicited”, not “the joint is never elicited”); Table 7 numbers all correct.
3.C	Heckman	Needs P17a (“no sample size repairs” is the paper’s inference); “nothing guarantees” is p. 109.
3.D	Bohren-Imas-Rosenberg	Needs P17b (attenuation along histories; $\tau_q \rightarrow 0$); pages are working-paper pages.
5.B	Propositions ORD and ADJ	Mathematics verified. Prose needs P10/P11: the one-sided construction is Hogarth-Einhorn’s Eq. 8; what is new is its two-attribute Jeffrey embedding. Needed for the omega defined in Setup (C.9). Optional Epstein (2006, eq. 12) citation.
6.A	Rival mechanisms	Needs P10 (Hogarth-Einhorn direction), P14 (six stimulus terms, eighteen response traits), P9 (Hawthorne’s objection reaches the two-basis case).
6.B	Two channels; rubric prediction	Carries the interior-omega numbers C.5 points to; position-channel wording needs the C.4 change.
B.A	Bibliography	Hawthorne2004 added; Heckman1998, Doring1999, Garber1980, Bohren2019 still to add when 3.A-3.D land; Jeffrey2004 and BenjaminBodohCreedRabin2019 if C.4/C.6/C.10 are approved.

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